

Weinberg-operator coefficient

$$T3 - C, \alpha = -1 : \quad S_1 = (2, -\frac{1}{2}), \quad S_2 = (2, \frac{1}{2}), \quad F = (3, 0)$$

$$C_5 = \frac{2}{3} \left(-\frac{\hbar M_F^3 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\hbar M_F^3 M_{S1}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} - \frac{\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \right)$$